

Redundancy Removal and Knowledge Alignment-Based Personalized Federated Learning for Online Condition Monitoring

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Abstract—Ensuring the safety and reliability of high-voltage electrical equipment requires robust and accurate online monitoring systems for detecting partial discharges (PDs) in switchgear. However, the transmission and storage of data in such systems raise significant concerns regarding data security and privacy. To address these issues, federated learning (FL) has gained popularity by enabling collaborative model training across clients without sharing raw data. Unlike previous approaches that compute model aggregation weights based solely on individual client performance, we propose a novel FL framework that evaluates client models using a strategy based on maximum diversity and minimum redundancy, resulting in a more informative and representative global model. Furthermore, a spatial-logic alignment module is introduced to enhance client model personalization for local data, with spatial alignment reinforced via a novel knowledge distillation mechanism. The proposed framework features a hybrid architecture composed of leaf clients, branch clients, and a central server. This structure allows branch clients to benefit from the aggregated global knowledge while preserving privacy among isolated clients. Leveraging edge computing at the client level ensures fast data processing and low-latency decision-making, thereby improving its monitoring capabilities. We validate our framework on a custom-built PD dataset and three public datasets. Experimental results demonstrate that our method outperforms the state-of-the-art approaches.

Index Terms—Knowledge distillation, online monitoring system, partial discharge (PD), personalized federated learning (PFL), redundancy removal (RR).

I. INTRODUCTION

THE development of robust and precise online monitoring systems for detecting partial discharges (PDs) in electrical switchgear is essential for ensuring the safety and reliability of high-voltage equipment. These systems are pivotal in enabling proactive maintenance strategies and preventing equipment failures by providing real-time insights into

the condition of critical assets [1], [2]. The integration of Internet of Things (IoT) technologies has further enhanced these systems, extending monitoring capabilities across all areas of power substations [3], [4]. However, effectively leveraging the vast amount of status information collected from these distributed substations remains a significant challenge. The primary issue lies in developing a reliable and accurate PD identification model that also preserves the confidentiality and integrity of sensitive data. Moreover, given the substantial value and sensitivity of this data, ensuring its security is also crucial. Thus, a comprehensive PD online monitoring system must excel not only in predictive accuracy but also in implementing strict data security measures to protect critical information.

Detecting PDs involves multiple challenges, such as handling noisy data, accommodating diverse PD patterns, and achieving accurate, real-time classification [5], [6]. Phase-resolved partial discharge (PRPD) patterns visually illustrate PD activity relative to a full 360° AC cycle over a designated period [7]. Recently, convolutional neural networks (CNNs) have gained high attraction for their ability to effectively extract key features from PD data, showing promising results in classifying various PD types [8]. However, few of these approaches are designed specifically to meet the demands of online monitoring and data privacy [9], [10]. For individual entities like laboratories, manufacturers, or power companies, gathering enough data samples is often difficult due to the high costs associated with specialized equipment and operations. Additionally, datasets from a single source may lack diversity. A potential solution is to combine data from various entities to train a unified model. However, issues such as conflicting interests and strict privacy requirements create significant obstacles to data sharing across organizations. As a result, implementing the centralized training approach is often unfeasible [11].

Unlike traditional centralized machine learning methods, federated learning (FL) offers a decentralized paradigm that enables collaborative model training across multiple edge devices or nodes while keeping data stored locally [11]. By allowing distributed clients to jointly optimize a shared model without exchanging raw data, FL effectively preserves data privacy throughout the entire training process [12], [13]. A foundational FL method, the FedAvg algorithm [14], adopts a client-server architecture to coordinate information exchange during training. In each communication round, the global

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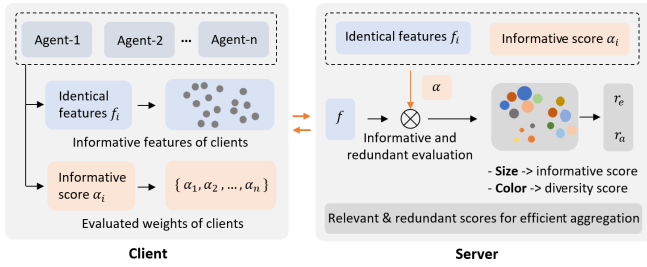


Fig. 1. Learned identification features, along with their evaluation scores, are utilized to compute the aggregation weights in the FL framework. This design aims to enhance the global model by maximizing knowledge diversity while minimizing redundancy, thereby achieving more efficient aggregation that accounts for both the informativeness and distinctiveness of client contributions.

model is distributed to participating clients for local updates. After local training, the server collects and aggregates the optimized parameters from all clients to refine the global model [15], [16]. By keeping sensitive data on local devices, FL minimizes communication overhead and reduces the risk of data leakage [17], [18], [19], [20], [21]. This decentralized learning approach is particularly advantageous for PD detection, where both data privacy and limited bandwidth are critical constraints. It enables edge devices deployed in power substations to collaboratively train intelligent diagnostic models using locally collected PD signals from diverse equipment types and operating conditions, achieving efficient and privacy-preserving model development in distributed monitoring environments [22].

Data in FL often comes from diverse sources such as devices, organizations, and user environments, leading to significant heterogeneity and distribution shifts [23], [24], [25], [26]. Consequently, client data are typically non-independent and identically distributed (non-*i.i.d.*), which makes effective model convergence challenging. To address this, the key objective is to adapt the model to each client's unique data distribution. Personalized federated learning (PFL) [27], [28], [29], [30], [31] has emerged as a promising direction, enabling clients to build personalized models better suited to local data rather than relying solely on a global model [32], [33], [34], [35], [36], [37]. However, PFL still faces several challenges. First, non-*i.i.d.* data can cause large divergence between local and global models, complicating the balance between personalization and global consistency. Second, limited data and computational resources at individual clients hinder the training of robust personalized models. Third, maintaining privacy and security while enabling flexible model adaptation remains difficult. Lastly, existing aggregation strategies often overlook the nuanced contributions of heterogeneous clients, reducing personalization performance. These challenges underscore the need for advanced PFL frameworks that can dynamically adapt to client diversity while preserving the advantages of collaborative learning.

In this study, we propose a novel PFL framework for online PD monitoring systems. As illustrated in Fig. 1, the framework incorporates a strategy based on maximizing diversity and minimizing redundancy to generate a more informative and

representative global model. This design effectively mitigates model divergence caused by non-*i.i.d.* data by encouraging complementary knowledge exchange among heterogeneous clients, thereby maintaining a balance between personalization and global consistency. Moreover, by adaptively weighting client contributions according to their informativeness, the framework enhances the utilization of limited data and computational resources on individual clients, enabling more robust and personalized local models.

In addition, the proposed approach ensures data privacy and security by keeping all raw PD data locally stored while exchanging only model-level information, thus minimizing the risk of information leakage. The diversity–redundancy optimization further allows the server to distinguish meaningful and nonoverlapping knowledge among clients, overcoming the limitations of conventional aggregation schemes that overlook client heterogeneity. Designed for deployment in challenging online monitoring environments, the proposed method achieves accurate, efficient, and privacy-preserving PD detection even under complex real-world conditions. Extensive experiments on both custom and public datasets, together with comparisons against state-of-the-art methods, demonstrate the effectiveness, robustness, and superiority of the proposed framework.

We summarize the main contributions as follows.

- 1) We have designed a novel redundancy removal (RR) module to reduce redundancy among features learned by different clients, thereby enhancing the quality of updates to the global model.
- 2) We have designed a novel knowledge alignment module to improve the performance of struggling clients and align the learned features across clients, leading to more accurate recognition.
- 3) We have implemented an online condition monitoring system that includes server and client models, integrating personalized FL with an informative knowledge learning scheme.
- 4) We have conducted ablation studies and comparisons using a real PD dataset and two public datasets to assess the effectiveness of the proposed method.

The rest of this article is organized as follows. Section II describes the related works from three aspects of edge-computing-based methods for online condition monitoring, data heterogeneity in FL system, and personalized federated learning methods. The preliminary knowledge is presented in Section III. In Section IV, we present the detailed design of our proposed PFL methodology for online condition monitoring. Section V presents the findings of our experiments. Finally, in Section VI, we conclude this work.

II. RELATED WORKS

A. Edge Clients for Online Condition Monitoring

Edge computing represents a revolutionary paradigm in computing, bringing computation and data storage closer to the sources or devices, thereby reducing dependence on centralized cloud servers [38], [39]. This paradigm offers several advantages, including real-time analysis, enhanced

data privacy, and the ability for local decision-making. In PD monitoring, edge devices are crucial for real-time data collection, processing, and analysis of PD signals, allowing for immediate insights and timely responses [38], [40]. Recent advancements include a MobileNet-based approach for recognizing PD patterns in gas-insulated switchgear (GIS), which improves accuracy and addresses challenges in training deep CNN models [41]. Additionally, a novel edge computing method introduced in [42] enhances PD detection by performing similarity computations after clustering sensor data, showing greater effectiveness and cost-efficiency. Zhao et al. [43] proposed an intelligent edge computing algorithm utilizing ultrasonic sensors and CNNs for early PD warning in switchgear.

The study by Klein et al. [44] explored edge devices for detecting PD originating from insulated conductors on high-voltage power lines, while the work in [45] investigated a stacked ensemble neural network for contactless PD classification. These studies highlight the fusion of edge computing and AI techniques, enabling precise PD recognition in resource-constrained environments. Techniques like transfer learning and knowledge distillation are used to transfer insights from complex models to more lightweight versions. The study in [46] describes a deep learning algorithm based on efficient CNNs, designed for real-time monitoring, fault diagnosis, and dynamic control on edge computing nodes. Furthermore, the study in [47] examines the architecture of deploying lightweight networks within an edge computing environment, facilitating direct deployment on resource-constrained edge devices.

B. Heterogeneous Data in Online Condition Monitoring

Traditional FL seeks to train a unified global model by aggregating knowledge from multiple clients without requiring them to share private data. The FedAvg algorithm [14] updates the global model by averaging models trained on each client's local data. This approach assumes that client data is independently and identically distributed (*i.i.d.*). However, the actual situation is often more complex and changeable. In real-world scenarios, client data is typically heterogeneous, and the data within each client usually follows a unique and non-*i.i.d.* distribution. Moreover, clients often differ in computational and communication resources, particularly in cross-device FL environments. These variations can significantly impact the performance of FedAvg in non-*i.i.d.* settings. To tackle these challenges, various approaches have been proposed. Many of these methods introduce penalties to client updates or adjust aggregation strategies at the server to alleviate the effects of data heterogeneity. For instance, methods like FedProx [48], pFedME [49], and FedDyn [50] use regularization terms to constrain client models, pulling them closer to the global model and reducing divergence caused by non-*i.i.d.* data. These strategies ensure that client updates remain aligned with the overall objective. Other methods, such as MOON [51], FCCL [52], FedUFO [53], FedProto [54], FPL [55], and FedProc [56], implement alignment-based penalties that standardize feature representations across clients, promoting

a more unified feature space across decentralized models to address data heterogeneity.

Additionally, approaches like SCAFFOLD [57] and FedDC [58] use gradient correction techniques to keep local updates aligned with the global gradient direction. Meanwhile, strategies in the works of [59], [60], [61], and [62] are designed to adjust aggregation weights dynamically during the global model update process. Though these methods improve upon FedAvg in managing non-*i.i.d.* data, they often struggle with extreme data heterogeneity. FedAS [63], however, employs a PFL approach that allows for client-specific model customization while maintaining global synchronization. By incorporating both local parameter alignment and client synchronization, FedAS provides a robust solution to the challenges posed by severe data heterogeneity. However, these methods typically rely solely on the evaluation metrics of individual client models to inform the fusion of the global model, without considering correlations or redundancies among multiple client models.

C. Personalized FL

FL has rapidly evolved with innovations addressing critical issues such as privacy, heterogeneity, non-*i.i.d.* data, and resource constraints. For instance, Li et al. [32] provide a comprehensive survey of threats and defenses across the FL lifecycle, highlighting the importance of trustworthy FL services. In terms of privacy protection, novel mechanisms such as hypernetwork-based parameter sharing [64] and privacy-enhanced edge device authentication [65] have been proposed. Attacks on vertical FL have also been explored in the form of spy attacks [66], while Jiang et al. [67] introduce an efficient model recovery scheme against poisoning attacks via certified rollback strategies. From a systems perspective, hierarchical and decentralized FL frameworks as in [20] and [21]. For model heterogeneity and robustness, representation-based aggregation and contrastive learning strategies are proposed in [68] and [69] to mitigate the effects of inconsistent model structures and corrupted data. Real-world deployments of FL are also being enhanced through semisupervised distillation pipelines, as demonstrated in [70]. The intersection of FL and quantum computing is explored in [19], providing a taxonomy for quantum FL.

Unlike traditional FL that trains a global model, PFL allows clients to retain personalized models that better fit their local data while still benefiting from shared knowledge [17], [18], [28], [71], [72]. An active client selection strategy using uncertainty-based metrics is introduced in [73] to improve both convergence speed and model accuracy with reduced communication overhead. In [74], a dual-module approach of FedPRP is proposed by combining federated personalization with prototype rectification. Similarly, methods like FedPer [27], FedBABU [75], and FedProto [54] promote personalization by enabling clients to share feature extractors while adapting classifiers. In [76], a novel framework of FedRoD is proposed by decoupling model learning into a generic predictor with robust losses for nonidentical class distributions and a lightweight personalized module to minimize each client's empirical risk. FedRep [77] and FedPAC [78] also improve local model

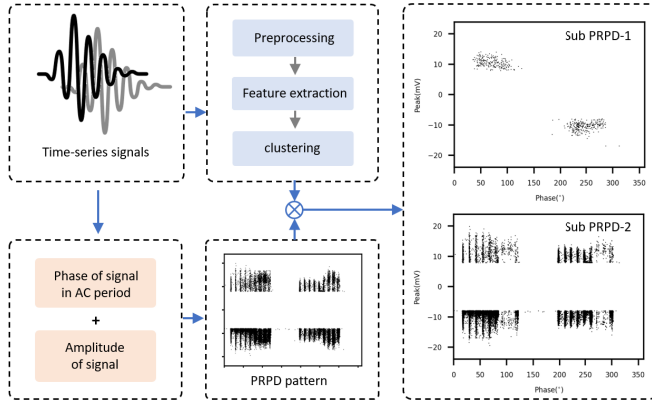


Fig. 2. Illustration of the proposed signal separation module. Sub-PRPD patterns are generated using a clustering algorithm that separates different types of signals based on features extracted from time-series waveforms. These sub-PRPD patterns are then adopted to PD classification.

alignment using client-specific prototypes. However, many PFL methods still face challenges such as poor alignment between shared and personalized components, inactive clients, and synchronization issues. To address these, asynchronous updates [11] and adaptive communication intervals [79] have been introduced, improving diagnostic accuracy and efficiency in real-time monitoring tasks. Edge-cloud collaboration [80] enables domain-adaptive models in complex environments, as seen in GIS fault diagnosis [22] and smart manufacturing [81].

III. PRELIMINARIES

A. PD Data Processing and Preparation

Fig. 2 gives out the procedure of processing the captured signals for PD classification. Due to the inherently random nature of PD events, detection systems must perform continuous signal acquisition to reliably capture meaningful PD signals as they occur. To facilitate effective analysis of the time-domain signals collected, we utilize the short-time Fourier transform (STFT) for spectral decomposition, allowing us to observe the frequency characteristics of PD signals over time. The spectral features obtained through STFT are represented as a feature set, $\mathbf{X}$, which is derived using (1). We further process $\mathbf{X}$ using kernel principal component analysis (KPCA)

$$\mathbf{X}(m, k) = \sum_{n=0}^{N-1} w(n-m) x(n) e^{-j2\pi nk/N} \quad (1)$$

where $x(n)$ is the n th value in time-serial signal, and k represents the frequency. $w(n)$ is the Hanning window function with length $N = 128$ and overlap is set to 64. m is the virtual-time index. $\mathbf{X}(m, k) \in \mathbb{R}^{d_t \times d_f}$ is obtained with balanced resolution in time and frequency domains, where d_t and d_f denote the length of time and frequency segments.

We typically utilize the Gaussian kernel function, transforming the initial feature $\mathbf{X}$ into a reduced space using $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-|\mathbf{x}_i - \mathbf{x}_j|^2 / 2\sigma^2)$, where $\mathbf{x}_i$ and $\mathbf{x}_j$ represent different feature vectors in $\mathbf{X}$, and $\sigma \in \mathbb{R}$ is the kernel's bandwidth parameter. The kernel matrix $\mathbf{K}$ is then centered as $\hat{\mathbf{K}} = \mathbf{K} - \mathbf{1}_N \mathbf{K} - \mathbf{K} \mathbf{1}_N + \mathbf{1}_N \mathbf{K} \mathbf{1}_N$, where $\mathbf{1}_N$ is an

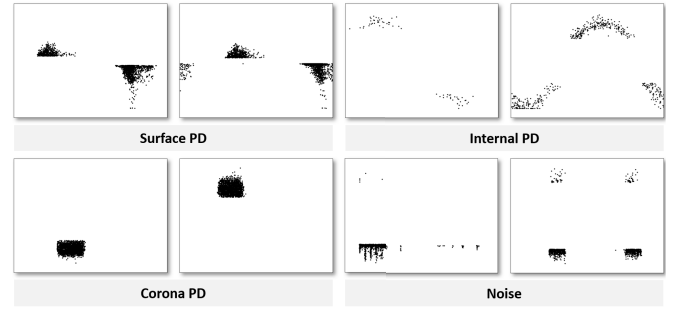


Fig. 3. Example images of the constructed PD dataset. Areas with darker colors represent that there are more signals within the area.

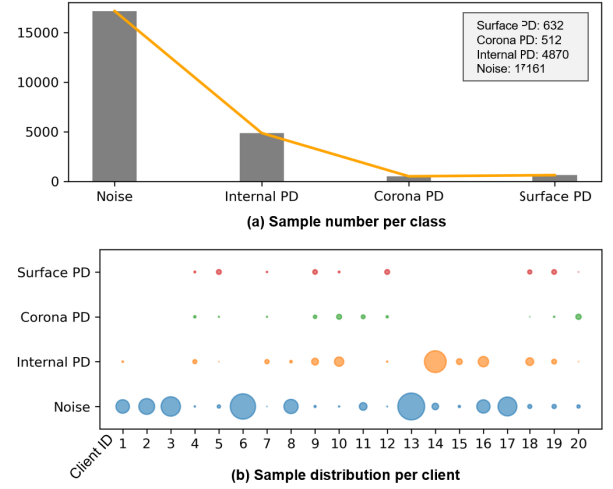


Fig. 4. (a) Sample distribution across different PD categories in our dataset. (b) Sample distribution across different clients, where larger circles indicate a greater number of samples assigned to each corresponding client.

$N \times N$ matrix with each element equal to $1/N$. Solving the eigenvalue problem $\hat{\mathbf{K}}\mathbf{v} = \lambda\mathbf{v}$ yields the eigenvalues λ and eigenvectors $\mathbf{v}$, allowing us to formulate the reduced feature $\mathbf{f}$ as $\sum_{j=1}^N a_j k(\mathbf{x}_i, \mathbf{x}_j)$, where a_j is the j th component of $\mathbf{v}$. Subsequently, we employ the k-means algorithm to differentiate PD signals from noise. To identify the optimal number of clusters, we use the Silhouette Score [82] as a measure of how well a data point fits within its cluster compared to others. Finally, these clustered PRPD patterns are normalized and formatted into 32×32 gray images for subsequent PD classification, as depicted in Fig. 3. Three types of PDs and noise signals are organized and annotated manually based on the actual collected data. The dataset consists of 23 175 PRPD patterns in total, and federated distribution is illustrated in Fig. 4. The size of the circle denotes the number of samples belonging to each category and client.

B. Personalized FL

PFL enhances traditional FL by addressing non-*i.i.d.* data across clients. In PFL, the model is divided into shared and personalized components to better accommodate the unique data distributions of each client. The shared module, denoted as θ_s , captures common knowledge accessible to all N clients, while each personalized module $\{\theta_{p,i}\}_{i=1}^N$ enables adaptation to

TABLE I
SIX-LAYER CNN NETWORK [63] FOR FL

Layer Type	Configuration
Conv2D Layer	Channels: 1 $\rightarrow$ 32, Kernel: 5x5, Stride: 1
ReLU	Inplace
MaxPool2D	Kernel: 2x2, Stride: 2
Conv2D Layer	Channels: 32 $\rightarrow$ 64, Kernel: 5x5, Stride: 1
ReLU	Inplace
MaxPool2D	Kernel: 2x2, Stride: 2
Flatten	-
FC Layer	In: 1600, Out: 512
ReLU	Inplace
FC Layer	In: 512, Out: Class number

the specific data characteristics of individual clients. Given N clients, where each client i has a unique local dataset $\mathcal{D}_i$, the model for client i is represented as $\Phi_i = (\theta_s, \theta_{p,i})$, comprising both shared and personalized modules. Let $\mathcal{L}_i(\Phi_i) := \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_i} [\mathcal{L}_i(\mathbf{x}, y; \theta_i)]$ denote the empirical loss for client i . The objective in PFL aims to optimize this loss function across all clients as follows:

$$\min_{\Phi_i \in \{\Phi_1, \dots, \Phi_N\}} \left\{ \frac{1}{N} \sum_{i=1}^N \mathcal{L}_i(\Phi_i) \right\} \quad (2)$$

where Φ_i denotes the i th client's model, and the aim is to achieve the minimum errors on the various data distributions located in different clients.

However, due to privacy requirements, transmitting raw data directly between clients and the central server is strictly prohibited. Instead, only model parameter updates, instead of the raw data, are shared to preserve data confidentiality. Let W_i^t represent the model parameters of client i at round t . After local training, client i updates its parameters to W_i^{t+1} with the change represented as ΔW_i . Each client then sends its update ΔW_i to the central server, where these updates are combined to produce the new shared parameter $\theta_s^{t+1} = \theta_s^t + 1/N \sum_{i=1}^N \Delta W_i$. The aggregated global model is subsequently distributed to all clients. This aggregation method aligns with the FedAvg approach [14], a fundamental FL technique that averages the contributions of participating clients to form an updated shared model. Considering the limited computational resources of edge-computing clients, we follow the setting in [63] to use a six-layer CNN model for all comparison methods in our experiments. The specific network structure is outlined in Table I. The settings of input and output of the model are adjusted according to different classification tasks.

IV. METHODOLOGY

In this work, we present the RR and knowledge alignment-based PFL scheme to handle the task of accurate online condition monitoring, as illustrated in Fig. 5. Following the overall framework of the designed scheme, Section IV-A details the knowledge alignment-based leaf client design. Section IV-B introduces the privacy-isolated branch module, while Section IV-C discusses the RR module developed for PFL.

A. Knowledge Alignment-Based Leaf Client

In PFL, each client's model is composed of two parts, denoted as $\Phi_i = (\theta_s, \theta_{p,i})$. The shared part, θ_s , is obtained from the server after aggregating updates from all clients. This shared model contains global knowledge, enabling it to produce more informative feature representations. Meanwhile, the personalized component, $\theta_{p,i}$, is trained locally within each client to classify local feature representations. However, there exists a significant inconsistency between the globally aggregated model θ_s and the locally trained model $\theta_{p,i}$, which leads to differing behaviors in processing local data. This discrepancy also challenges the ability of the local head to accurately classify representations from other clients. To overcome this challenge, our goal is to incorporate local knowledge into the shared parameters, enabling them to produce representations that are consistent with the local classifier. Specifically, we use the previous local-tuned model θ_s^{t-1} as guidance to optimize the current model θ_s^t , aligning their outputs to achieve consistency with the personalized head.

To enhance knowledge alignment between the global and client models, we introduce spatial maps derived from the CNN feature maps, as illustrated in Fig. 6. These spatial maps capture the spatial distribution of activations, thereby complementing conventional feature-based knowledge distillation. By incorporating spatial information, the proposed module transfers not only semantic representations but also structural spatial knowledge, effectively mitigating knowledge forgetting during model adaptation in FL. Unlike previous methods that focus solely on aligning the knowledge between θ_s^t and θ_s^{t-1} in the feature representation domain, the proposed approach also emphasizes spatial alignment to obtain more informative and consistent knowledge. The knowledge from the previously locally tuned model is preserved in the extracted spatial map $\mathbf{M}^{t-1} \in \mathbb{R}^{w \times h}$ and the feature representation $\mathbf{v} \in \mathbb{R}^d$, where w and h denote the width and height of the spatial map, and d represents the dimension of the feature vector. Accordingly, we denote these two types of information obtained by current round of global model θ_s^t as $\mathbf{M}^t \in \mathbb{R}^{w \times h}$ and $\mathbf{z} \in \mathbb{R}^d$, respectively. We then transfer the informative knowledge within $\mathbf{M}^{t-1}$ and $\mathbf{v}$ to the global model θ_s^t using the following equation:

$$\mathcal{L}_{al} = \mathbb{E} \left[(\hat{\mathbf{v}}^T \hat{\mathbf{z}} - 1)^2 + \lambda_{al} \|\mathbf{M}^t - \mathbf{M}^{t-1}\|_2 \right] \quad (3)$$

where $\mathbf{v}$ and $\mathbf{M}^{t-1}$ are the learned feature representation and spatial map by the previous local-tuned model θ_s^{t-1} . For the same input of x , the two results of $\mathbf{z}$ and $\mathbf{M}^t$ are extracted by the current round global model θ_s^t . The two features are normalized by $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ and $\hat{\mathbf{z}} = \mathbf{z}/\|\mathbf{z}\|$. Therefore, we have $\|\hat{\mathbf{z}}\| = 1$. The gradient of $\mathcal{L}_{al}$ respect to $\hat{\mathbf{z}}$ is calculated to pull feature $\hat{\mathbf{z}}$ toward the direction of $\hat{\mathbf{v}}$.

In addition to the feature representations, spatial information is also aligned with the previous round model by minimizing the term of $0 \leftarrow \|\mathbf{M}^t - \mathbf{M}^{t-1}\|_2$. Follow this, we calculate two types of distance losses to optimize the global model θ_s^t , learn features with high relevance to the previous round model. As to the spatial information, we want them to be

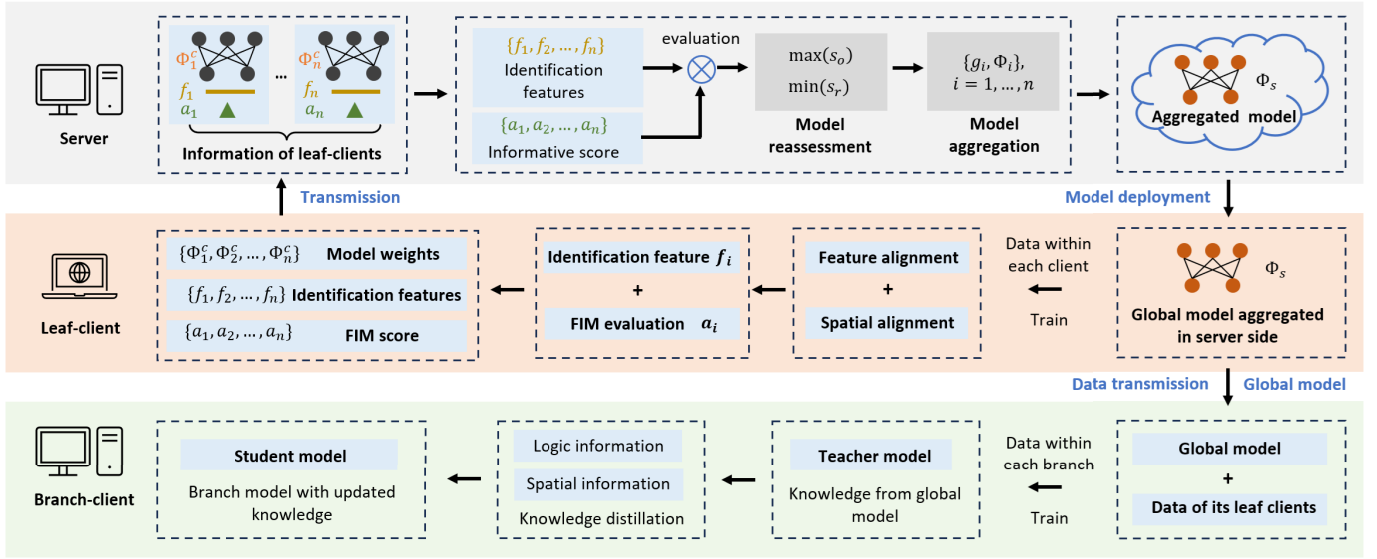


Fig. 5. Illustration of the proposed RR and knowledge alignment-based PFL scheme. The entire system is structured into three levels: server, branch-client, and edge-client. The trained models at the edge-client side are transmitted to the server for global model aggregation by the proposed RR and knowledge alignment modules. At the branch-client side, a privacy-isolated model is deployed to learn knowledge from the aggregated model from the server to improve its performance.

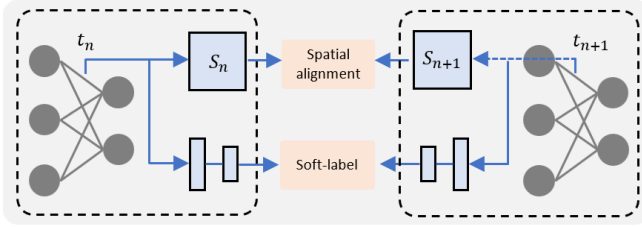


Fig. 6. Illustration of the spatial-logic alignment module for personalized knowledge learning on edge clients. The aggregated global model is further fine-tuned using personalized knowledge from the previous timestamped model.

the same in the spatial dimension by calculating the l_2 -norm of the interpolation between two spatial maps. Here, l_2 -norm penalizes large discrepancies more strongly, ensuring that both direction and magnitude shifts are minimized across rounds. This results in more stable convergence and better alignment across model updates, especially in real-world federated settings with significant client heterogeneity. In this way, the converged features learned by the global model at the latest round will be aligned with the previous round of the model, eliminating the inconsistency between the global model and local personalized models.

B. Privacy Isolated-Based Branch Client

In certain applications, specific clients' data may be excluded from shared model updates to ensure strict data privacy and protect intellectual property. To address this, a privacy-isolated model is implemented with a branch-client established between the shared server and these edge-computing clients, as shown in Fig. 5. This branch-client processes data for clients requiring high security or unique intellectual property protection. To support this architecture,

we employ a spatial-logical knowledge distillation model, which transfers knowledge between the globally trained model θ_s (aggregated from distributed clients) and the branch-client model $\theta_{b,i}$. This setup results in a fault detection model that maintains privacy isolation while improving performance.

It should be noted that the global model θ_s is used as the teacher model to enhance the generalization of branch-client models θ_b by improving accuracy and upholding privacy. To achieve this, we further adopt the soft labels as the guidance to enhance the robustness of the branch client's classifier. The insights acquired by the global model are valuable assets that can boost the efficacy of the branch models and aid them in acquiring more insightful knowledge. This knowledge can then be converted into a format of soft-label as defined in the following equation:

$$\hat{p}_i = \frac{e^{(\hat{o}_i/T)}}{\sum_k^c e^{(\hat{o}_k/T)}} \quad (4)$$

where $\hat{o}_i$ signifies the logical output of the global federated model for the true class i . The temperature parameter of T controls the calculated soft-label, and c is the class count. When $T < 1$, the probability distribution of the distilled output $\hat{p}_i$ becomes sharper, while it becomes broader when T increases. Consequently, the ability to recognize different types of PDs learned by the well-trained global model θ_g (denoted as teacher model) can be transmitted to the branch-client model θ_b (denoted as student model) through knowledge distillation.

To emphasize the importance of spatial information for the PD classification model, an innovative spatial-logic information-based knowledge distillation module is introduced. The loss function for this item can be defined in the

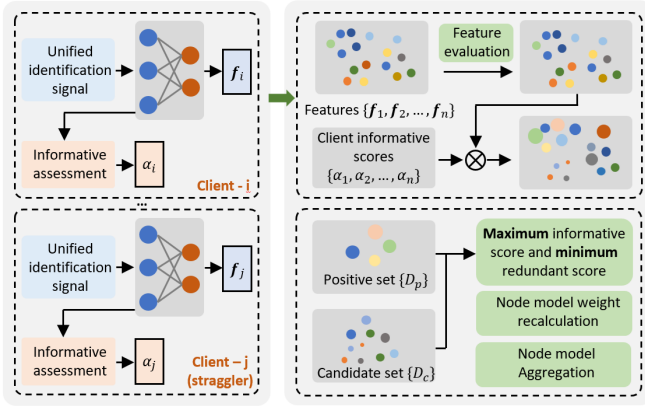


Fig. 7. Illustration of the proposed strategy for server-side model aggregation based on maximum informative diversity and minimum redundancy. Identification features and evaluation scores from each edge client are used to compute the aggregation weights, resulting in a global model that maximizes informative diversity while minimizing redundancy.

following equation:

$$\mathcal{L}_{kd} = -\frac{1}{C} \sum_{i=1}^C (y_i + \hat{p}_i) \log \hat{y}_i + \lambda_{kd} \|\mathbf{M}_s - \mathbf{M}_b\|_2 \quad (5)$$

where $\hat{p}_i$ is the distilled results for class i . y_i and $\hat{y}_i$ signify the one-hot label and predicted result for true class i . $\mathbf{M}_s$ and $\mathbf{M}_b$ denote the spatial results obtained by the global federated model and branch-client model, respectively. λ_{kd} is the balancing parameter.

C. RR for PFL

Fig. 7 illustrates the design for evaluating both the effectiveness and redundancy of client models. The proposed strategy aims to aggregate client models in a way that maximizes informative knowledge while minimizing redundancy. In the following content, we detail and discuss these two aspects.

1) *Informativeness Evaluation*: In non-i.i.d. scenarios, a global model formatted by averaging the weights of clients experiences a significant performance drop without considering the various data distributions within the clients. Therefore, proper weights for each client's model should be assigned after obtaining various models trained by different clients using their local data. Follow this, we first adopt the score in [63] to evaluate the models trained by different clients, which indicate the ability of the locally trained model Φ_i in each client. The Fisher information matrix (FIM) is calculated in the following equation:

$$\text{FIM} = \nabla_{W_i} \log p(D_i|W_i) \cdot \nabla_{W_i} \log p(D_i|W_i)^T \quad (6)$$

$$\alpha_i = \sum_{j=1}^{d_w} \text{diag}(\text{FIM})_j \quad (7)$$

where $\nabla_{W_i} \log p(D_i|W_i)$ denotes the gradient for i th client's model parameters W_i . The private dataset for client i is denoted as D_i . Additionally, α_i is computed as the trace of the Fisher information matrix (t-FIM), capturing the informativeness of the client's learned features. Given the high computational cost of second-order derivatives and the limited processing

capabilities of edge-computing devices in practice, we use the sum of the diagonal elements as a proxy to estimate the knowledge contained in a client model. Each client computes its α_i value and transmits it to the server, where these values are normalized for aggregation weighting by $\alpha_i = \alpha_i / \sum_{i=1}^N \alpha_i$. Generally, larger value of α_i represents more informative knowledge have been learned by the i th client. Accordingly, a higher weight will be assigned during model aggregation for these distributed clients.

2) *Redundancy Evaluation*: However, we should not only look at the ability of the model to learn knowledge, but also address potential knowledge redundancy among client models during aggregation. In other words, we hope that the aggregated model has the maximum knowledge efficiency and diversity while minimizing the redundancy across the models during knowledge aggregation. To achieve this, an identification feature is introduced to describe the learned knowledge by each client's model. We assume that the well-trained model has gained informative knowledge from its local data. Therefore, the extracted deep features using the latest model can represent the knowledge of its local data, which can be used to represent the learned knowledge of each client. Specifically, this feature representation is the mean feature vector extracted from each client's local data (512-dimension as defined in Table I). Given two identification feature vectors, $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{\mu} = [\mu_1, \mu_2, \dots, \mu_n]$ with length of n , kernel density estimation (KDE) is further used to estimate their marginal and joint probability distributions. The marginal probability density $\hat{P}(x)$ can be computed as

$$\hat{P}(x) = \frac{1}{\sqrt{2\pi n} \sigma h} \sum_{i=1}^n e^{k(x, x_i)} \quad (8)$$

where $k(x, x_i)$ is the function of $-(x - x_i)^2 / 2h^2$, and h is the smoothing parameter that determines the width of the kernel. In this way, the marginal probability distributions $\hat{P}(\mathbf{v})$ and $\hat{P}(\mathbf{\mu})$ for feature representations of $\mathbf{v}$ and $\mathbf{\mu}$ can be calculated with above (8).

To quantify the relationship between the variables presented in two identification features of $\mathbf{v}$ and $\mathbf{\mu}$, the joint probability distribution of $\mathbf{v}$ and $\mathbf{\mu}$ is defined as

$$\hat{P}(x, y) = \frac{1}{2\pi n h_x h_y} \sum_{i=1}^n e^{k(x, x_i)} e^{k(y, y_i)} \quad (9)$$

where $\hat{P}(x, y)$ is the estimated joint probability density at x and y . Two functions of $k(x, x_i)$ and $k(y, y_i)$ are denoted as $-(x - x_i)^2 / 2h_x^2$ and $-(y - y_i)^2 / 2h_y^2$, respectively. h_x and h_y are the smoothing parameters that determine the width of the kernel. The joint probability density function $\hat{P}(x, y)$ captures the relationship between two feature dimensions simultaneously, while the marginal probability density function $\hat{P}(x)$ reflects the distributions of each feature independently. Mutual information quantifies the informational dependency between the two features by comparing the divergence between their joint probability distribution and the product of their marginal probability distributions.

Algorithm 1 Algorithm for Model Aggregation Based on Maximum Diversity of Informative Knowledge and Minimum Redundancy

Input: Identification feature vectors $\{\nu_1, \nu_2, \dots, \nu_N\}$, corresponding evaluation scores $\{\alpha_1, \alpha_2, \dots, \alpha_N\}$, balancing parameter λ_{rr} .

Output: Weights $\{g_i\}$ for model aggregation.

- 1 Initialize $S \leftarrow \{\nu_k\}$ where $\nu_k = \arg \max_i^N \alpha_i$;
- 2 Set weight $g_k \leftarrow \alpha_k$;
- 3 Initialize remaining feature set $R \leftarrow \{\nu_i : i \neq k\}$;
- 4 **while** R is not empty **do**
- 5 **for each** $\nu_i \in R$ **do**
- 6 Compute diversity gain
 $r(\nu_i, S) = \frac{1}{|S|} \sum_{\nu_j \in S} D(\nu_i, \nu_j)$;
- 7 Normalize $r(\nu_i) = \sigma(r(\nu_i, S))$;
- 8 Compute total gain
 $g(\alpha_i, \nu_i) = \lambda_{rr} \alpha_i - (1 - \lambda_{rr}) r(\nu_i)$;
- 9 Select the model $\nu_{\text{best}} \leftarrow \arg \max_{\nu_i \in R} g(\alpha_i, \nu_i)$;
- 10 Add ν_{best} to the selected set S : $S \leftarrow S \cup \{\nu_{\text{best}}\}$;
- 11 Remove ν_{best} from R : $R \leftarrow R \setminus \{\nu_{\text{best}}\}$;
- 12 Set weight $g_{\text{best}} \leftarrow g(\nu_{\text{best}})$;
- 13 **return** $\{g_i\}$ where $g_i = g_i / \sum_{i=1}^N g_i$;

To measure the redundancy between ν and μ , mutual information is further calculated based on the probability distributions

$$I(\nu; \mu) = \frac{1}{n} \sum_{i=1}^n \log \frac{\hat{P}(\nu_i, \mu_i)}{\hat{P}(\nu_i) \hat{P}(\mu_i)} \quad (10)$$

where $\hat{P}(\nu)$ and $\hat{P}(\mu)$ are the marginal probability density. $\hat{P}(\nu, \mu)$ is the estimated joint probability density. A higher mutual information value of $I(\nu; \mu)$ indicates a stronger dependency between the two features ν and μ .

3) *Convergence Analysis:* We consider a set of models $\{\Phi_1, \Phi_2, \dots, \Phi_N\}$ within various clients, and each can be represented by its extracted identification feature vector ν and associated evaluation score α . Let $\{\nu_1, \nu_2, \dots, \nu_N\}$ be the feature set and $\{\alpha_1, \alpha_2, \dots, \alpha_N\}$ be the score set. Our goal is to select a subset of models that not only have high evaluation scores but also maximize feature diversity. However, selecting the best subset of models to ensure highly informative knowledge diversity and low redundancy is challenging. We hence adopt the greedy algorithm to progressively find a balance between the evaluation score of models and the diversity of their extracted features. The motivation behind adopting a greedy algorithm stems from its simplicity and ability to yield good approximations for optimization problems, particularly when the objective is a combination of performance, diversity, and redundancy. As depicted in Algorithm 1, selecting the optimal set S that satisfies the maximum evaluation score and diversity as well as the minimum redundancy. We introduce λ_{rr} to balance the two parts for each client candidate before model aggregation. Then, models within various clients $\{\theta_i\}_{i=1}^N$ are aggregated together to format the global model according to the calculated score by $\sum_{i=1}^N g_i \theta_i$.

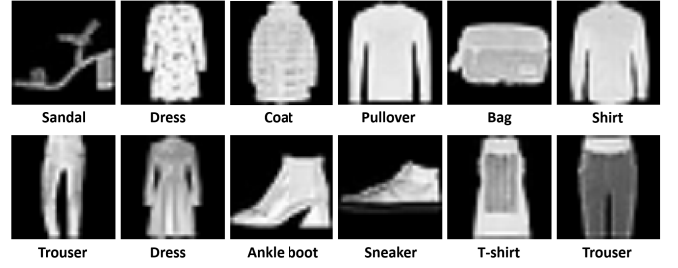


Fig. 8. Examples image of the Fashion-MNIST dataset. Each image has a size of 28×28 .

The algorithm converges to a fixed set of weights g_i within at most N iterations, determined by the bounded score range and the well-defined gain function. This convergence is guaranteed due to the finite size of the problem and the monotonic nature of the selection process. Specifically, termination is ensured because the while loop in Algorithm 1 runs at most N times, transferring one knowledge vector from the candidate set R to the selected set S in each iteration. Monotonic improvement is achieved through the selection of ν_{best} at each step, which maximizes the objective function $g(\alpha_i, \nu_i)$ by choosing the candidate ν_i with the highest value. The function $g(\alpha_i, \nu_i)$ is designed to balance informativeness α_i and diversity $r(\nu_i, S)$, thereby ensuring progressive optimization. Once the candidate pool R is exhausted, the algorithm terminates, with S representing the optimal selection under the given criteria. The iterative maximization of $r(\nu_i, S)$ ensures convergence to a stable set of weights g_i , which are subsequently normalized to satisfy $\sum g_i = 1$, yielding a consistent and interpretable weighting scheme.

V. EXPERIMENTAL RESULTS

In this section, we begin by providing a comprehensive description of our experimental setup. We then proceed to conduct a detailed analysis of the effectiveness of our proposed techniques, which include RR module and spatial-logic knowledge alignment (SLKA) module for PFL. We conduct a comparison between the proposed PFL framework (FedR-RKA) and some state-of-the-art methods. Additionally, we analyze the performance of these models on the constructed PD dataset with different FL settings. We also conducted comparative experiments on three public datasets of Fashion-MNIST [83], CIFAR10 [84], and CIFAR100 [84] to verify the generalization of the proposed method. Comparisons with several state-of-the-art FL strategies are conducted to demonstrate the effectiveness of the proposed approach.

A. PD Dataset and Implementation Details

1) *Datasets:* We collect thousands of real PD signals from multiple power substations to construct a comprehensive PD dataset. Three types of PDs, along with noise signals, are manually annotated based on the actual collected data. The final dataset comprises 23 175 PRPD image patterns. Notably, the dataset exhibits significant class imbalance, with noise and internal PD categories containing substantially more samples than the other two categories. For additional evaluation, we



Fig. 9. Examples image of the CIFAR10/100 dataset. Each image has a size of 32×32 .

adopt three public datasets. Among them, Fashion-MNIST [83] is a collection of Zalando's article images as shown in Fig. 8, consisting of 70 000 gray scale images sized 28×28 pixels. Each image is labeled with one of 10 classes. CIFAR10 and CIFAR100 [84] include 60 000 color images of size 32×32 pixels, as shown in Fig. 9, distributed across 10 and 100 classes, respectively.

For these three datasets, samples are divided using a Dirichlet distribution [85] and distributed among the clients. The Dirichlet distribution's β parameter, set at $\{0.1, 0.3, 0.5\}$, allows us to simulate different degrees of non-*i.i.d.* distribution and analyzes the impact on accuracy. Generally, distributed subsets across various clients under larger β values represent more non-*i.i.d.* distributions. We also varied the client online ratios to simulate the struggling clients in real applications, denoted as P . Lower value of P indicates that more straggling clients exist. This experimental configuration simulates real-world scenarios and enables a comprehensive evaluation of our method.

2) *Counterparts*: We compare our approach with several advanced methods including FedAS [63], FedALA [86], FedPer [27], FedRep [77], GPFL [87], and FedDBE [88]. Experimental results on the PD dataset and the public datasets achieved by these methods are evaluated. It's worth noting that traditional FL algorithms, such as FedAvg [14] and FedProx [48], are primarily focused on producing a generalized global model without considering the scenarios of PFL. To enable a fair comparison, we further fine-tune the aggregated global models generated by FedAvg and FedProx to create personalized versions, reported as FedAvg-P (Personalized) and FedProx-P.

3) *Implementation Details*: For fair comparison, we adopt identical hyperparameters across all four datasets. In the CIFAR10 [84], CIFAR100 [84], and Fashion-MNIST [83] experiments, samples are randomly split into 50% for training, 25% for validation, and 25% for testing. For the highly imbalanced PD dataset, we address class imbalance by selecting 200 samples per category for the test set to ensure a more accurate evaluation of classification performance across different PD types. PRPD patterns are converted to gray scale image format to facilitate effective model training. To enhance robustness, data augmentation techniques such as random phase shifts and vertical flips are applied to simulate variations in sensor orientation and initialization effects. In our FL scheme, 20 clients participate in training, each performing 5 local epochs within a communication round. The global model

TABLE II
EXPERIMENTS ON THE VALIDATION SET OF CIFAR100 DATASET WITH DIFFERENT SETTINGS OF THE RR MODULE

Setting	$\beta=0.1, P=0.2, \lambda_{al}=0$					
λ_{rr}	1.0	0.9	0.8	0.7	0.6	0.5
Acc.	52.38%	52.67%	52.81%	52.71%	52.54%	52.22%

is updated through 40 communication rounds during model aggregation. Local optimization is conducted using the stochastic gradient descent (SGD) algorithm with a learning rate of $1e^{-2}$, a decay rate of $1e^{-3}$, and a batch size of 16. The random seed for PyTorch and NumPy is fixed at 0 to ensure reproducibility of all experimental results.

B. Model Ablation Studies

1) *Evaluation of PD Signal-Noise Separation*: Classifying and identifying the original PRPD pattern directly is often influenced by noise interference. However, using our proposed signal-noise separation module, we can decompose this mixed signal source into distinct and individual ones. This allows for more precise analysis and identification of the collected signals. As depicted in Fig. 10, two sets of collected signals are clustered into different groups. The sub-PRPD patterns corresponding to the clustering results are generated with more clear and accurate PD patterns. It can be noted that the PDs are separated from the noises, as in the sub-PRPD patterns with red color. Compared with the raw PRPD patterns, this can reduce the impact of noise signals on the PRPD pattern of PD signals, thereby affecting the results of the PD recognition model. To quantify the effectiveness of the noise separation, we adopt the metric C_{IoU} to assess the module's performance in clustering the signals. Specifically, C_{IoU} is defined as $\max_{i=1, \dots, m} (S_{pd} \cap S_i) / (S_{pd} \cup S_i)$, where S_{pd} is the set of PD signals based on ground truth labels, m represents the number of clusters, and S_i corresponds to the i th cluster group. This measures the overlap between the clusters and the actual PD group S_{pd} . On our dataset, we achieved $C_{IoU} = 0.969$. A higher C_{IoU} value, approaching 1, indicates successful separation of PD signals from noise, reflecting superior performance of the signal-noise separation module.

2) *Evaluation of RR Module*: To assess the proposed RR module, we conduct an analysis from two perspectives: correlation between the features and model evaluation scores, and classification performance of the aggregated model on different clients. Thus, the relations among extracted identification features are estimated, which provides us with richer indicators for evaluating models of different clients. If we simply assign weights to the models based on the evaluation scores, it will often reduce the diversity of information contained in the models. By comprehensively evaluating these two indicators, the aggregated model can have better performance. As demonstrated in Table II regarding the results on the validation set of the CIFAR100 dataset, the proposed RR module maintains a balance between the informative and diversity of the learned knowledge for model aggregation. It achieves a margin of

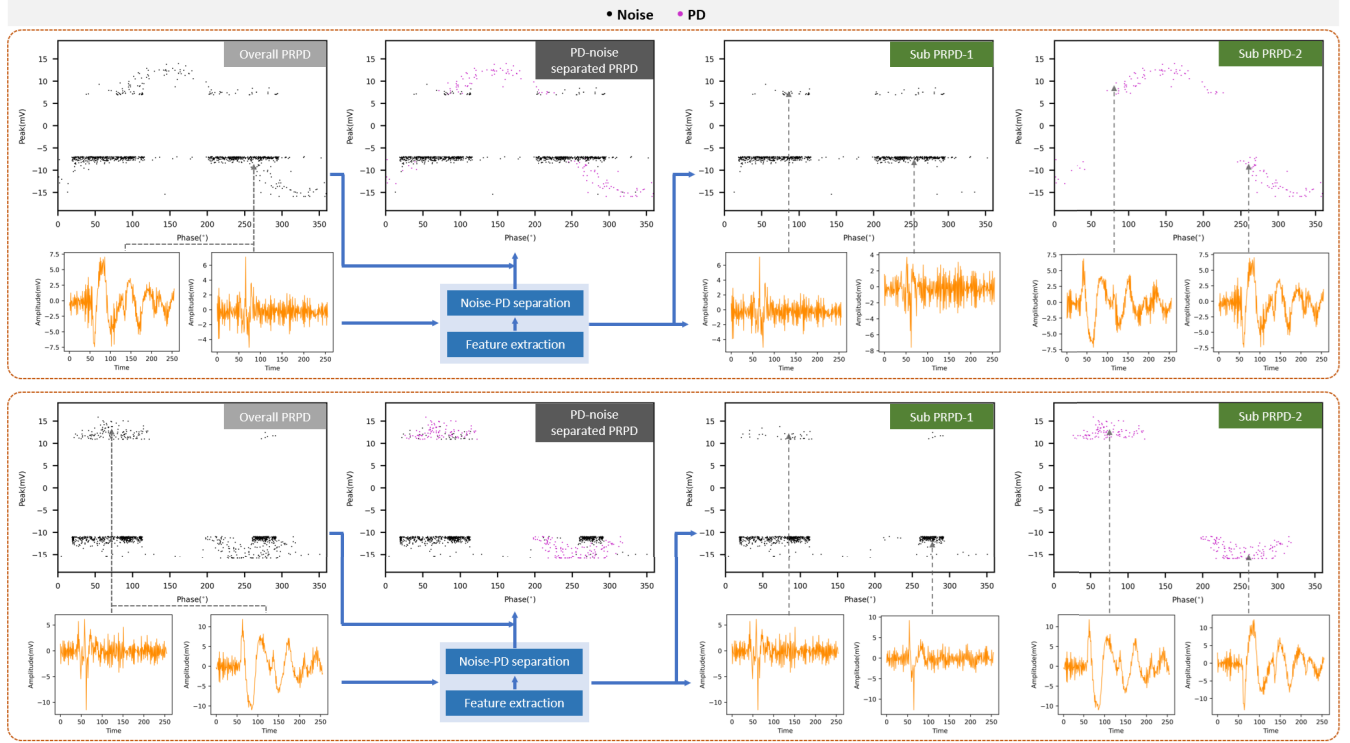


Fig. 10. Signal separation results for two collected signal sets. Clustering results are applied to the overall PRPD patterns to generate sub-PRPD patterns, and each sub-PRPD pattern can be regarded as signals from a single source.

TABLE III
EXPERIMENTS ON THE VALIDATION SET OF CIFAR100 DATASET
WITH DIFFERENT SETTINGS OF THE SLKA

Setting	$\beta=0.1, P=0.2, \lambda_{rr}=1.0$						
λ_{al}	0	0.2	0.4	0.6	0.8	1.0	1.2
Acc.	52.38%	52.56%	52.64%	52.73%	52.68%	52.61%	52.47%

0.43% compared with the model aggregated by only relying on the evaluation score.

3) *Evaluation of Knowledge Alignment Module*: Knowledge alignment module is used to align the learned features with previous ones to obtain personalized knowledge for handling local data within each client. To evaluate this, we conduct experiments on the PD dataset with different settings for the spatial-logic alignment module. In detail, the aggregated global model by the server will be used to evaluate on the testing sets by each client directly when the knowledge alignment module is disabled. In this condition, that PFL model conducting without the knowledge alignment module is some kind of traditional FL, e.g., the FedAvg and FedProx. Table III showcases that the proposed method provides marginal performance enhancements of 52.73%. Conversely, the model achieves better generalization for the classifier and reaches higher accuracy.

4) *Evaluation of Privacy-Isolated Scheme*: Experiments are also conducted to evaluate the effectiveness of the proposed privacy-isolated scheme. We set up separate experiments to verify the performance improvement of the proposed module

on these three datasets. To be specific, two clients are further selected as the privacy-isolated clients. Performance of the branch-client is analyzed when adopting the spatial-logic knowledge distillation to the hybrid FL module or not. The parameter of β in Dirichlet distribution and participate rate P is set to 0.1 and 0.2. Follow this, the branch-client achieves the baseline accuracy of 91.3%, 88.4%, and 50.1% on the clients' local data of the PD, CIFAR10, and CIFAR100 datasets without relying on the knowledge of the aggregated global model. Then, the knowledge distillation module is enabled and setting the weight of λ_{kd} to 0.3. The proposed spatial-logic knowledge distillation module is adopted to transfer the informative and generalized knowledge to the branch-client. With the help of the knowledge distillation module, large margins of 2.5%, 1.3%, and 1.8% are obtained on the three datasets.

C. Experimental Results Analysis

1) *PD Dataset*: After obtaining the best settings for the knowledge alignment and RR modules, we combine the training and validation sets to train our PFL model. The training and testing loss together with the accuracy curves for the PD dataset are shown in Fig. 11. To reflect practical conditions and thoroughly assess our model's robustness and compatibility, we partition the data using a Dirichlet distribution [85] and distribute it across all clients. We adjust the β parameter in the Dirichlet distribution to simulate varying levels of non-*i.i.d.* data distribution. In our experiments, we set β to values {0.1, 0.3, 0.5} to represent different degrees of data heterogeneity. That is, larger β values indicate the subsets

TABLE IV
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON THE CONSTRUCTED PD DATASET WITH
DIFFERENT SETTINGS OF β AND PARTICIPATE RATE P

Methods	$\beta = 0.1$				$\beta = 0.3$				$\beta = 0.5$			
	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$
FedAvg-P [14]	92.88	95.75	97.63	98.00	92.75	95.75	97.25	97.50	91.63	94.88	95.75	96.25
FedProx-P [48]	92.63	95.88	97.50	98.25	92.88	95.13	97.38	97.38	91.38	94.50	95.63	96.00
FedDBE [88]	92.75	95.13	97.25	97.50	92.50	95.25	96.38	96.63	91.75	95.13	96.25	96.38
GPFL [87]	92.88	95.75	97.63	97.88	93.63	95.50	97.13	97.25	91.88	95.38	96.13	96.88
FedPer [89]	93.50	96.25	97.63	98.13	93.25	96.38	97.00	97.50	92.50	95.75	96.25	96.75
FedRep [77]	93.63	96.38	97.25	97.63	93.75	95.88	97.13	97.38	91.38	96.13	96.88	96.50
FedALA [86]	93.88	96.63	97.75	98.25	94.13	96.75	97.63	97.88	93.88	96.38	97.00	96.88
FedAS [63]	94.63	97.25	98.13	98.50	94.38	96.75	97.88	98.00	94.25	96.88	97.13	97.25
FedRRKA(Ours)	95.13	97.63	98.50	98.75	95.00	97.25	98.13	98.38	94.63	97.38	97.25	97.50
Δ	$\uparrow 0.50$	$\uparrow 0.38$	$\uparrow 0.37$	$\uparrow 0.25$	$\uparrow 0.62$	$\uparrow 0.50$	$\uparrow 0.25$	$\uparrow 0.38$	$\uparrow 0.28$	$\uparrow 0.50$	$\uparrow 0.12$	$\uparrow 0.25$

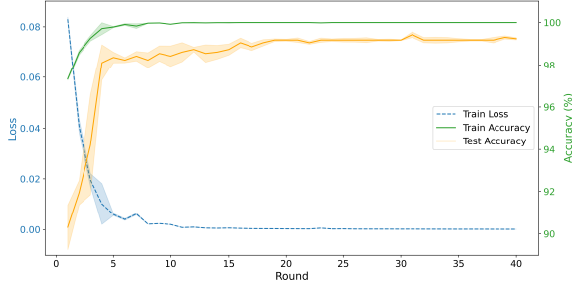


Fig. 11. Curves of the loss and accuracy on the testing set of PD dataset achieved by the proposed model.

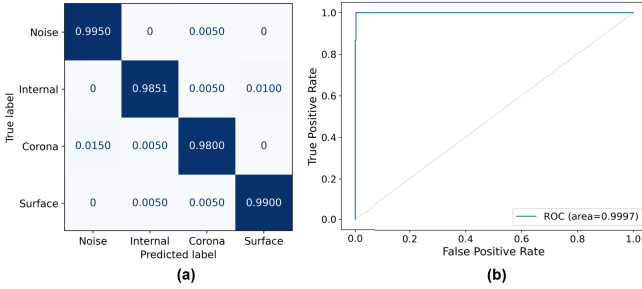


Fig. 12. ROC and PR curves of the proposed method on the testing set of PD dataset are illustrated in (a) and (b), respectively.

within each client have greater non-*i.i.d.* distribution. Client online ratios, denoted as P , are also varied in $\{0.2, 0.4, 0.6, 1\}$. More straggling clients are found during the FL if the P is set to smaller values.

As shown in Table IV, our proposed method, denoted as FedRRKA, achieves the highest accuracy across all settings. The experimental results demonstrate its strength in personalization and in decreasing the effect of inactive clients. Additionally, we compute the confusion matrix for both the baseline and the proposed FedRRKA models to illustrate the enhanced ability to recognize different PD types. As displayed in Fig. 12, the accuracy of identifying various PD types is improved, leading to lower miss detection and false alarm rates in the online condition monitoring system.

2) *FMNIST Dataset*: We also conduct experiments on the dataset of Fashion-MNIST to further evaluate the generalizability and robustness of our proposed method. Similar to the other datasets, we also adjust the β parameter

in the Dirichlet distribution to simulate varying levels of non-*i.i.d.* data distribution. The parameter of β is set to $\{0.1, 0.3, 0.5\}$ for simulating different degrees of data heterogeneity. As demonstrated in Table V, our method achieves the best results for most scenarios compared with these baseline methods. However, the results of the two scenarios are slightly lower than FedALA [86]. This is because when the knowledge differences among clients are small, meaning each client has already achieved relatively good performance, the effectiveness of our aggregation strategy brings little room for further performance improvement. However, in most FL scenarios where data heterogeneity is severe, the performance differences among clients are significant, and our strategy offers a clear advantage in such cases. The optimized aggregation weights for obtaining the global model contribute to the robust performance of the proposed FL scheme.

3) *CIFAR10 and CIFAR100 Datasets*: We further evaluate the effectiveness of our proposed method on two public datasets of CIFAR10 and CIFAR100. For both datasets, data is partitioned using a Dirichlet distribution [85] and allocated across all clients following the same configuration as the PD and Fashion-MNIST datasets. By adjusting the β parameter in the Dirichlet distribution, we simulate varying degrees of non-*i.i.d.* conditions to assess model performance under different levels of data heterogeneity. This experimental design closely reflects real-world FL scenarios, enabling a comprehensive evaluation of the robustness and adaptability of our method. As presented in Tables VI and VII, the results include comparisons with several state-of-the-art approaches, encompassing both aggregation-based and feature engineering-based strategies. Notably, our proposed method achieves the highest accuracy on both datasets. In particular, under higher values of β , where data heterogeneity is more pronounced, FedRRKA demonstrates a significant performance advantage over competing methods.

D. Computational and Communication Cost Analysis

We summarize the comparison and analysis of the computational and communication costs, as demonstrated in Table VIII. For the client side, the model used at each client is a lightweight CNN, consisting of two convolutional layers and two fully connected layers. Given the input data with size

TABLE V
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON THE FASHION-MNIST DATASET WITH DIFFERENT SETTINGS OF β AND PARTICIPATE RATE P

Methods	$\beta = 0.1$				$\beta = 0.3$				$\beta = 0.5$			
	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$
FedAvg-P [14]	77.11	80.57	81.80	82.47	88.42	88.58	89.13	88.74	88.96	89.78	89.80	90.08
FedProx-P [48]	81.06	81.20	83.02	82.91	88.27	88.58	89.16	89.07	89.38	89.07	90.03	89.93
FedDBE [88]	83.43	84.15	85.39	86.99	88.99	89.94	91.05	91.40	89.93	90.49	91.15	91.18
GPFL [87]	93.14	93.54	95.39	93.79	91.36	90.38	90.24	89.40	91.16	91.25	91.28	91.48
FedALA [86]	97.16	97.12	97.29	97.31	93.91	94.19	94.32	94.36	92.50	92.83	92.71	93.02
FedRep [77]	97.21	97.23	97.35	97.34	93.11	93.66	93.78	93.86	90.95	91.64	92.16	92.23
FedPer [89]	97.25	97.35	97.35	96.42	93.93	93.82	93.98	93.96	92.19	92.16	92.23	92.19
FedAS [63]	97.32	97.47	97.53	97.48	93.61	94.17	94.31	94.29	92.33	92.67	92.65	92.57
FedRRKA(Ours)	97.45	97.68	97.75	97.72	94.08	94.55	94.47	94.49	92.56	92.89	92.81	92.72
Δ	$\uparrow 0.13$	$\uparrow 0.21$	$\uparrow 0.22$	$\uparrow 0.24$	$\uparrow 0.15$	$\uparrow 0.36$	$\uparrow 0.15$	$\uparrow 0.13$	$\uparrow 0.06$	$\uparrow 0.06$	$\uparrow 0.10$	$\downarrow 0.30$

TABLE VI
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON CIFAR10 DATASET WITH DIFFERENT SETTINGS OF β AND PARTICIPATE RATE P

Methods	$\beta = 0.1$				$\beta = 0.3$				$\beta = 0.5$			
	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$
FedAvg-P [14]	56.95	67.04	78.71	85.26	57.71	68.26	73.64	76.60	63.47	68.21	72.15	73.59
FedProx-P [48]	56.84	66.03	78.04	84.67	57.39	67.50	73.09	76.18	62.14	67.45	71.54	72.86
FedDBE [88]	56.89	57.16	63.25	66.55	62.96	65.01	67.71	69.46	63.62	67.25	69.28	70.64
GPFL [87]	82.36	84.41	84.73	86.38	70.13	73.93	75.10	74.60	68.90	71.35	70.76	69.32
FedPer [89]	88.51	86.00	87.26	87.86	66.58	71.26	73.58	74.97	60.41	62.60	65.23	66.80
FedRep [77]	89.48	90.18	90.32	90.52	79.94	80.30	80.76	80.72	72.50	77.47	74.55	74.47
FedALA [86]	89.61	89.89	90.00	90.06	79.03	80.92	81.02	81.40	75.05	76.24	76.52	76.75
FedAS [63]	89.92	90.60	90.81	91.03	80.23	81.58	82.01	82.12	75.93	77.29	77.35	77.97
FedRRKA(Ours)	90.36	91.24	91.36	91.42	80.96	82.11	82.62	82.57	76.62	77.85	77.79	78.27
Δ	$\uparrow 0.44$	$\uparrow 0.64$	$\uparrow 0.55$	$\uparrow 0.39$	$\uparrow 0.73$	$\uparrow 0.53$	$\uparrow 0.61$	$\uparrow 0.45$	$\uparrow 0.69$	$\uparrow 0.56$	$\uparrow 0.44$	$\uparrow 0.30$

TABLE VII
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON CIFAR100 DATASET WITH DIFFERENT SETTINGS OF β AND PARTICIPATE RATE P

Methods	$\beta = 0.1$				$\beta = 0.3$				$\beta = 0.5$			
	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$	$P=0.2$	$P=0.4$	$P=0.6$	$P=1$
FedAvg-P [14]	27.61	36.42	41.61	47.57	27.87	32.67	36.35	38.67	27.44	30.66	32.07	33.82
FedProx-P [48]	27.42	35.40	40.41	45.96	27.27	32.25	35.29	37.85	26.99	30.22	31.36	33.46
FedDBE [88]	25.91	28.60	29.42	30.92	27.09	29.09	30.37	31.22	27.02	29.29	30.27	30.72
GPFL [87]	30.06	34.86	36.85	39.13	26.79	29.84	30.72	32.19	34.71	34.78	34.02	33.28
FedPer [89]	44.96	45.12	44.32	44.40	34.78	34.32	33.66	33.08	29.77	28.71	27.79	27.17
FedALA [86]	45.52	48.40	50.67	51.06	36.54	40.41	41.20	42.08	31.19	35.07	36.85	37.15
FedRep [77]	47.72	49.28	50.17	49.96	36.09	37.86	36.97	36.93	29.83	30.42	30.35	30.16
FedAS [63]	52.41	55.20	55.76	56.37	40.68	43.51	44.47	44.91	35.48	38.21	38.70	39.57
FedRRKA(Ours)	52.95	55.71	56.16	56.73	41.24	44.11	44.81	45.20	36.23	38.80	39.17	40.01
Δ	$\uparrow 0.54$	$\uparrow 0.51$	$\uparrow 0.40$	$\uparrow 0.36$	$\uparrow 0.56$	$\uparrow 0.60$	$\uparrow 0.34$	$\uparrow 0.29$	$\uparrow 0.75$	$\uparrow 0.59$	$\uparrow 0.47$	$\uparrow 0.44$

TABLE VIII
COMPARISON OF COMPUTATIONAL AND COMMUNICATION COST FOR THE BASELINE AND THE PROPOSED METHODOLOGY

Metric	Baseline	FedRRKA
Forward FLOPs	$\sim 15.65\text{M}$	$\sim 15.65\text{M}$
FIM computation	Yes	Yes
Mutual Info calculation	No	Yes ($\mathcal{O}(N^2d)$)
Greedy calculation	No	Yes ($\mathcal{O}(N^2)$)
Model size	$\sim 3.35\text{ MB}$	$\sim 3.35\text{ MB}$
Extra transmit	No	512-d vector
Communication load	$\sim 3.35\text{ MB}$	$\sim 3.35\text{ MB} + 2\text{ KB}$

of 32×32 , the estimated computational cost per forward pass is about 15.65 MFLOPs. This makes the model efficient for edge deployment. Compared with the baseline method, extra

transmit involves a 512-dimensional vector for global model aggregation. During training, backward operations approximately triple the cost, keeping it well within edge device capability. For the server side, while the baseline method uses only local FIM computation (based on output gradients), the proposed FedRRKA introduces an additional RR step. This involves calculating pairwise mutual information across clients and executing a greedy selection algorithm for model aggregation.

E. Drawbacks and Future Perspectives

The proposed FedRRKA framework effectively enhances the informativeness of the global model through optimized

server-side aggregation. However, several limitations remain for further investigation. This work does not comprehensively address privacy-preserving concerns or model robustness against adversarial attacks. Although differential privacy is employed, advanced threats such as model poisoning or gradient leakage are not fully considered and may pose significant risks in practical deployments. Additionally, as the scale of online monitoring systems expands, the increasing computational and communication costs on resource-constrained edge devices become critical bottlenecks that need more efficient solutions. The current design is also primarily evaluated on image classification tasks, which may limit its adaptability to other data types commonly encountered in industrial settings, such as time-series sensor data or multimodal inputs. Future research should explore the integration of stronger privacy-preserving mechanisms, enhanced scalability, and support for continual learning. All the above approaches are essential to ensure the robustness and broad applicability of FL in real-world industrial monitoring environments.

VI. CONCLUSION

In this study, we propose a novel approach for PFL designed for online PD monitoring systems, which integrates both informative knowledge and model diversity into the aggregation process. By leveraging the theory of information entropy, our scheme assigns more comprehensive and representative weights to client models, leading to improved aggregation quality. Additionally, we design a knowledge alignment module that combines spatial and logical information through knowledge distillation, enabling each client to obtain personalized knowledge while still benefiting from the global model. To further enhance knowledge transfer, we adopt a teacher–student framework that allows the well-trained federated model to distill valuable insights to privacy-isolated branch clients. Extensive experiments and comparisons with existing state-of-the-art methods demonstrate the effectiveness of the proposed framework.

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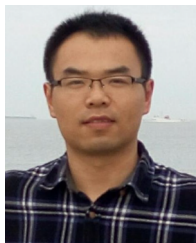


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